

Lab_Forma1_1

You have to check the design equivalence of the generated netlist post synthesis Verses RTL Verilog files using Formal Verification Techniques using Formality Tool GUI.

Steps: -

1. Copy Lab_Forma1_1 folder from your computer into the virtual machine inside Labs directory
2. Open syn_script.tcl
 - Add the following command under section “Formality Setup File” to generate svf file used in formal verification

- set_svf ALU.svf

```
6 ##### Formality Setup file #####
7
8 set_svf ALU.svf
9
10 ##### Design Compiler Library Files #setup #####
11
12 puts "#####"
13 puts "#      #setting Design Libraries      #"
14 puts "#####"
15
16 #Add the path of the libraries to the search_path variable
17 lappend search_path /home/IC/Labs/Lab_Forma1_1/rtl
18 lappend search_path /home/IC/Labs/Lab_Forma1_1/std_cells
19
```

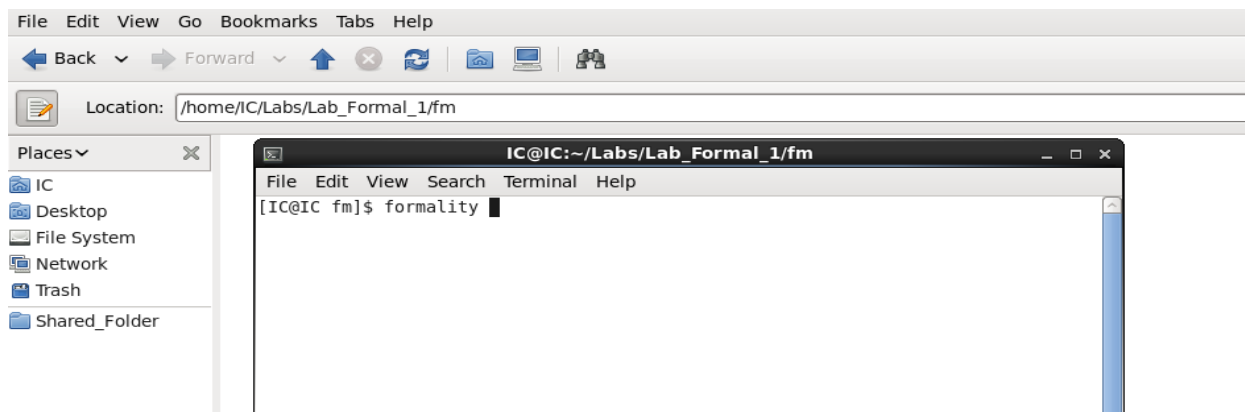
- Add the following command before write command under section “Close Formality Setup File” to stop writing in svf file and close it

```
62
63 ##### Close Formality Setup file #####
64
65 set_svf -off
66
67 #####
68 # Write out Design after initial compile
69 #####
70
71 write_file -format verilog -hierarchy -output ALU.v
72
73 ##### starting graphical user interface #####
74
```

3. Run synthesis using the following command to generate formality setup file “SVF File”
 - dc_shell -f syn_script.tcl | tee syn.log

4. Move to fm folder and then run the following command in the terminal to open formality tool GUI

- **formality**



5. Load Guidance Step

- Check “**Use Auto Setup**” checkbox.
- Load Guidance File (SVF) generated from synthesis tool

6. Load Reference Container

- Read Verilog Design Files
- Read DB Libraries
- Set Top Design

7. Load Implementation Container

- Read netlist File generated from synthesis
- Read DB Libraries
- Set Top Design

8. Run Match

9. Run Verify